

# SDS8102: Advanced Mathematical Statistics

## Lecture 1: Introduction to Statistical Thinking

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# Learning Objectives

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By the end of this lecture you should be able to

1. Explain the role of probability models in statistical inference.
2. Distinguish between populations and samples.
3. Define a statistical model mathematically.
4. Differentiate parametric and nonparametric models.
5. Recognize exponential and location-scale families.
6. Understand why statistical models are abstractions rather than reality.

# Why Do We Need Statistical Models?

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Suppose we wish to answer one of the following questions.

- Does a new drug improve patient survival?
- Is one machine learning model more accurate than another?
- Will tomorrow's stock market increase?

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Different questions, but they all share one common feature.

Uncertainty

Mathematical Statistics provides the rigorous theoretical foundation for analyzing data and measure uncertainty, test hypotheses, and make reliable decisions across scientific domains.

# The Fundamental Problem of Statistics

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Reality is never directly observable.

Instead we observe only a finite amount of data.

$$X_1, X_2, \dots, X_n$$

A **population** represents the complete collection of objects under study.

A **sample** consists of the observations that are actually collected.



**Challenge:** Conclusions about the population must be drawn using only the sample.

# Statistics in Modern AI

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Many AI systems are statistical procedures.

Examples include

- Large Language Models → Next token predictions.
- Medical diagnosis → Tumor classification.
- Recommendation systems → Markov decision process for song recommendations in Spotify.
- Self-driving vehicles → Next action (stop/ drive, turn left/right) prediction.
- Scientific discovery → Faster simulations for engineering designs.

Although the algorithms are different, they all rely on probabilistic reasoning from data.

# Statistical Models

Mathematical Statistics

# What is a Statistical Model?

## Definition: Statistical Model

A **statistical model** is a family of probability distributions

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\},$$

defined on a common measurable space  $(\mathcal{X}, \mathcal{A})$ , where

- $\mathcal{X}$  is the sample space,
- $\mathcal{A}$  is a  $\sigma$ -algebra,
- $\Theta$  is the parameter space (**may not be finite dimensional**),
- $\theta$  is the unknown parameter.



# Anatomy of a Statistical Model

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Every statistical model consists of four mathematical objects.

| Object        | Interpretation                         |
|---------------|----------------------------------------|
| $\mathcal{X}$ | Possible observations                  |
| $\mathcal{A}$ | Events whose probabilities are defined |
| $\Theta$      | Unknown parameter space                |
| $P_\theta$    | Probability law generating the data    |

The objective of statistical inference is to learn the unknown parameter  $\theta$  from observations generated according to  $P_\theta$ .

## Example: Coin Tossing

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Suppose a coin is tossed independently  $n$  times.

Each observation satisfies 
$$X_i = \begin{cases} 1, & \text{Head} \\ 0, & \text{Tail.} \end{cases}$$

The probability model is

$$X_i \sim \text{Bernoulli}(p), \quad 0 \leq p \leq 1$$

Here,

- Data:  $X_1, \dots, X_n$
- Parameter:  $p$
- Parameter space:  $\Theta = [0, 1]$

## Example: Heights of Students

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Suppose  $X_i =$  height of student  $i$ .

A common model is

$$X_i \stackrel{iid}{\sim} N(\mu, \sigma^2).$$

Unknown parameters are

$$\theta = (\mu, \sigma^2).$$

Hence  $\Theta = \mathbb{R} \times (0, \infty)$ .

# Models Are Approximations

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George Box famously wrote

**Remark:**

“All models are wrong, but some are useful.”

A statistical model is

- not reality,
- but a simplified mathematical description of reality.

The quality of our conclusions depends on how reasonable the model assumptions are.

# The Parameter Space

The parameter space contains every possible value of the unknown parameter.

Examples

| Model        | Parameter         | Parameter Space                 |
|--------------|-------------------|---------------------------------|
| Bernoulli    | $p$               | $[0, 1]$                        |
| Normal Mean  | $\mu$             | $\mathbb{R}$                    |
| Normal Model | $(\mu, \sigma^2)$ | $\mathbb{R} \times (0, \infty)$ |
| Poisson      | $\lambda$         | $(0, \infty)$                   |

# The Big Picture

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Throughout this course we will answer four fundamental questions.

1. How should we estimate unknown parameters?
2. How can we quantify uncertainty?
3. How do we compare competing hypotheses?
4. What happens when the model is incorrect?

These questions form the backbone of modern mathematical statistics.

# Choosing a Statistical Model

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We now know what a statistical model is.

The next question is equally important.

Which family of probability distributions should we use?

There are two broad approaches.

## Next Topic

- Parametric Models
- Nonparametric Models

## Parametric Models

A statistical model is **parametric** if it can be indexed by a finite-dimensional parameter.

### Definition: Parametric Model

A model

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

is parametric if the parameter  $\theta$  is finite-dimensional; typically,

$$\Theta \subseteq \mathbb{R}^d$$

for some finite  $d$ .

The dimension  $d$  does not depend on the sample size  $n$ .



## Example: Bernoulli Model

---

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Then

$$P_p(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

The model is

$$\mathcal{P} = \{P_p : p \in [0, 1]\} \implies \boxed{\dim(\Theta) = 1}$$

## Example: Normal Location-Scale Model

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2).$$

The parameter is  $\theta = (\mu, \sigma^2)$

with the parameter space

$$\Theta = \mathbb{R} \times (0, \infty) \implies \dim(\Theta) = 2.$$

The two coordinates have distinct statistical roles:

$$\mu = \text{location}, \quad \sigma = \text{scale}.$$

## Why Use Parametric Models?

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Parametric models impose structure on the data-generating mechanism.

This structure can provide

- low-dimensional summaries of the population,
- interpretable parameters,
- efficient estimation,
- tractable likelihoods,
- exact finite-sample results.

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- exact finite-sample results.

But there is a price:

|                                            |
|--------------------------------------------|
| More structure $\implies$ More assumptions |
|--------------------------------------------|

# The Risk of Parametric Assumptions

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Suppose the true distribution is  $P_0$ .

We decide to work with

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

There are two possibilities.

1.  $P_0 \in \mathcal{P}$  : The model is correctly specified.
2.  $P_0 \notin \mathcal{P}$  : The model is misspecified.

This distinction will become important when we study **M-estimation and quasi-likelihoods** in Week 2.

## A Natural Question

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How much structure do we really need?

Suppose we observe

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P.$$

Instead of assuming  $P = P_\theta$ , for some finite-dimensional  $\theta$ , why not allow

$$P \in \mathcal{P}$$

to range over a much larger class?

This leads us to **nonparametric models**.

# Nonparametric Models

A nonparametric model allows the unknown distribution to vary over a large, typically infinite-dimensional, class.

## Definition: Nonparametric Model

A model

$$\mathcal{P} = \{P : P \in \mathcal{F}\}$$

is called nonparametric when the unknown object indexing the model is infinite-dimensional.

## Example:

$\mathcal{P} = \{P : P \text{ is a probability distribution on } \mathbb{R}\}.$

## Parametric versus Nonparametric

|                   | Parametric                | Nonparametric      |
|-------------------|---------------------------|--------------------|
| Unknown object    | $\theta \in \mathbb{R}^d$ | $P$ or $F$         |
| Dimension         | Fixed finite $d$          | Typically infinite |
| Assumptions       | Stronger                  | Weaker             |
| Flexibility       | Limited                   | High               |
| Classical example | $N(\mu, \sigma^2)$        | Arbitrary $F$      |



# The Distribution Function as a Parameter

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For a real-valued random variable  $X$ , define

$$F(x) = P(X \leq x).$$

If we make no parametric assumption on  $F$ , then the unknown object is

$$F : \mathbb{R} \rightarrow [0, 1]. \quad (\text{infinite-dimensional parameter})$$

Thus, in a nonparametric model,

the parameter itself may be a function.

# The Empirical Distribution

---

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F.$$

A natural estimator of  $F$  is

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}.$$

This is the **empirical distribution function** (EDF).

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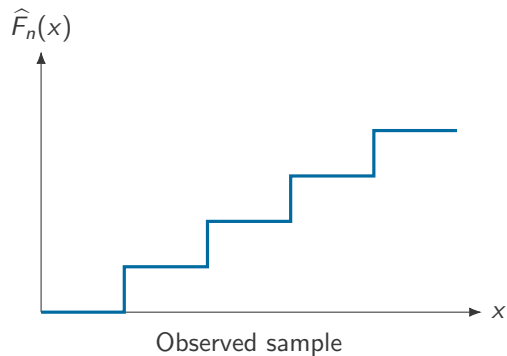
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Notice that no parametric form for  $F$  was required.

## A Geometric View



The empirical distribution places mass  $1/n$  at each observation.

## Why Nonparametric Models?

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Nonparametric models are attractive when a strong parametric assumption is difficult to justify.

They can provide

- greater modeling flexibility,
- robustness against distributional misspecification,
- direct access to features of the unknown distribution.

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- greater modeling flexibility,
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- direct access to features of the unknown distribution.

But this flexibility comes with a cost.

|                                            |
|--------------------------------------------|
| Less structure $\implies$ Harder inference |
|--------------------------------------------|

## Think About It

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Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F$$

and we want to estimate  $F$ .

Which approach uses more information?

1. Assume  $F = N(\mu, \sigma^2)$  and estimate  $(\mu, \sigma^2)$ .
2. Make no parametric assumption and estimate  $F$  directly.

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2. Make no parametric assumption and estimate  $F$  directly.

Neither is universally better.

The right choice depends on the scientific problem.



## Why Exponential Families?

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We have seen many parametric models:

Bernoulli, Poisson, Normal, Gamma, ...

A natural question is:

*Do these models share a common mathematical structure?*

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A natural question is:

*Do these models share a common mathematical structure?*

**YES !!**

Exponential families

provide one of the most important unifying structures in statistical inference.

## Definition: Exponential Family

### Definition: Exponential Family

A family of distributions with density

$$f_{\eta}(x) = h(x) \exp \{ \eta^{\top} T(x) - A(\eta) \},$$

with respect to a common dominating measure  $\nu$ , is an **exponential family**.

Here

- $\eta \in \mathcal{H} \subseteq \mathbb{R}^k$  is the **natural parameter**,
- $T(x) \in \mathbb{R}^k$  is the **natural statistic**,
- $A(\eta)$  is the **log-partition function**,
- $h(x)$  is the base measure term.

## A Useful Equivalent Form

An exponential family is often written as

$$f_{\theta}(x) = h(x) \exp \left\{ \sum_{j=1}^k \eta_j(\theta) T_j(x) - A(\eta(\theta)) \right\}.$$

The distinction between  $\theta$  and  $\eta$  is important.

$\theta$  = statistical parameter

whereas

$\eta$  = natural parameter.

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$\theta$  = statistical parameter

whereas

$\eta$  = natural parameter.

The natural parameterization often reveals the structure of the model much more clearly.

## Example: Bernoulli Family

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Let

$$X \sim \text{Bernoulli}(p).$$

Then

$$f_p(x) = p^x(1-p)^{1-x}.$$

Taking logarithms,

$$\log f_p(x) = x \log p + (1-x) \log(1-p).$$

Therefore,

$$f_p(x) = (1-p) \exp \left\{ x \log \frac{p}{1-p} \right\}.$$

## Bernoulli: Canonical Form

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Using

$$p = \frac{e^\eta}{1 + e^\eta},$$

we obtain

$$f_\eta(x) = \exp \{ \eta x - \log(1 + e^\eta) \}.$$

Thus

$$T(x) = x, \quad A(\eta) = \log(1 + e^\eta).$$

The Bernoulli family is therefore a one-parameter exponential family.

## Example: Poisson Family

Let  $X \sim \text{Poisson}(\lambda)$ . Then  $f_\lambda(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ ,  $x \in \mathbb{N} \cup \{0\}$ .

Rewrite:

$$f_\lambda(x) = \frac{1}{x!} \exp \{x \log \lambda - \lambda\}.$$

Set

$$\eta = \log \lambda.$$

Then

$$\boxed{f_\eta(x) = \frac{1}{x!} \exp \{\eta x - e^\eta\}} \implies T(x) = x, \quad A(\eta) = e^\eta.$$



## Example: Normal Family

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Let

$$X \sim N(\mu, \sigma^2).$$

Its density is

$$f_{\mu, \sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}.$$

Expand the square:

$$-\frac{(x - \mu)^2}{2\sigma^2} = -\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} - \frac{\mu^2}{2\sigma^2}.$$

Thus

$$f_{\mu, \sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ \frac{\mu}{\sigma^2} x - \frac{1}{2\sigma^2} x^2 - \frac{\mu^2}{2\sigma^2} \right\}.$$

## Normal Family: Natural Parameters

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Define

$$\eta_1 = \frac{\mu}{\sigma^2}, \quad \eta_2 = -\frac{1}{2\sigma^2}.$$

Then

$$T(x) = \begin{pmatrix} x \\ x^2 \end{pmatrix}.$$

The density takes the form

$$f_{\eta}(x) = h(x) \exp \{ \eta_1 x + \eta_2 x^2 - A(\eta) \}.$$

Thus the two-dimensional Normal family is an exponential family with

$$T(X) = (X, X^2)^{\top}.$$

## The Log-Partition Function

The function  $A(\eta)$  is determined by normalization:

$$\int h(x) \exp \{ \eta^\top T(x) - A(\eta) \} d\nu(x) = 1.$$

Therefore,

$$e^{A(\eta)} = \int h(x) e^{\eta^\top T(x)} d\nu(x).$$

Hence

$$A(\eta) = \log \int h(x) e^{\eta^\top T(x)} d\nu(x).$$

This is why  $A$  is also called the **log-partition function**.

## Why Is $A(\eta)$ Important?

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The derivatives of  $A$  encode moments of the natural statistic.

Under regularity conditions,

$$\nabla A(\eta) = \mathbb{E}_{\eta}[T(X)].$$

Moreover,

$$\nabla^2 A(\eta) = \text{Var}_{\eta}(T(X)).$$

Therefore,

$$\boxed{\nabla^2 A(\eta) \succeq 0}$$

and  $A$  is convex.

The exponential family therefore connects **probability** and **convex analysis**.

# Exponential Families and Samples

Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f_\eta.$$

Then the joint density is

$$\prod_{i=1}^n f_\eta(x_i).$$

Using the exponential-family representation,

$$L(\eta; x_{1:n}) = \left[ \prod_{i=1}^n h(x_i) \right] \exp \left\{ \eta^\top \sum_{i=1}^n T(x_i) - nA(\eta) \right\}.$$

Notice what happened to the data:

## A First Look at Sufficiency

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The likelihood depends on the sample through

$$T_n(X_1, \dots, X_n) = \sum_{i=1}^n T(X_i).$$

This suggests that

$$T_n = \sum_{i=1}^n T(X_i)$$

contains all the information about  $\eta$  needed for likelihood-based inference.

This is our first encounter with the idea of a **sufficient statistic**. (Next Lecture)

# The Exponential-Family Structure

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$$f_{\eta}(x) = h(x) \exp\{\eta^{\top} T(x) - A(\eta)\}$$

This single representation gives us:

- a finite-dimensional natural statistic  $T(X)$ ,
- a tractable likelihood,
- moment identities through derivatives of  $A$ ,
- a direct route to sufficient statistics,
- useful convexity and geometric structure.

**These properties will drive much of our later theory.**

# Location Families

Let  $f$  be a density on  $\mathbb{R}$ .

A **location family** is a collection of densities of the form

$$f_{\mu}(x) = f(x - \mu), \quad \mu \in \mathbb{R}.$$

The parameter  $\mu$  shifts the distribution without changing its shape.

$$\boxed{f(x) \xrightarrow{\text{shift by } \mu} f(x - \mu)}$$

Examples

$$N(\mu, \sigma^2), \quad \text{Cauchy}(\mu, \gamma), \quad \text{Logistic}(\mu, s).$$



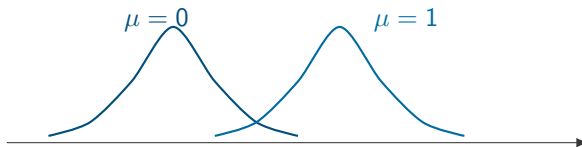
## Location Families: Interpretation

Suppose

$$X = \mu + \varepsilon,$$

where the distribution of  $\varepsilon$  does not depend on  $\mu$ .

Then  $X - \mu = \varepsilon$ . Thus changing  $\mu$  simply translates the distribution.



## Scale Families

---

Let  $f$  be a density on  $\mathbb{R}$ .

A **scale family** is a collection of densities

$$f_{\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right), \quad \sigma > 0.$$

The parameter  $\sigma$  changes the scale or spread of the distribution.

## Scale Families

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A **scale family** is a collection of densities

$$f_{\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x}{\sigma}\right), \quad \sigma > 0.$$

The parameter  $\sigma$  changes the scale or spread of the distribution.

If  $X = \sigma Z$ ,

then

$$\frac{X}{\sigma} = Z.$$

Thus changing  $\sigma$  stretches or contracts the distribution.

## Location-Scale Families

Combining the two transformations gives a **location-scale family**:

$$f_{\mu,\sigma}(x) = \frac{1}{\sigma} f\left(\frac{x - \mu}{\sigma}\right), \quad \sigma > 0.$$

Equivalently,

$$X = \mu + \sigma Z,$$

where the distribution of  $Z$  is fixed.

Location controls **where** the distribution is.

Scale controls **how spread out** it is.

## Example: The Normal Location-Scale Family

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Let  $Z \sim N(0, 1)$

Then

$$X = \mu + \sigma Z$$

has distribution

$$X \sim N(\mu, \sigma^2).$$

Its density is

$$f_{\mu, \sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2 \right\}.$$

The Normal family is therefore a location-scale family.

# Why Location-Scale Families Matter

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Location-scale families provide a simple way to separate

shape

location

scale.

Many statistical procedures are invariant under transformations

$$X \mapsto a + bX, \quad b > 0.$$

This perspective will become useful when we study

- properties of estimators,
- hypothesis tests,
- confidence regions,
- optimality and decision theory.

# Lecture 1: What Have We Built?

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We began with the question:

How do we mathematically represent uncertainty?

We introduced

Statistical Models

and distinguished

Parametric      vs.      Nonparametric.

We then identified two important parametric structures:

Exponential Families

Location-Scale Families.

## The Road Ahead

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A statistical model gives us a family of possible distributions.

But the data are still

$$X_1, \dots, X_n.$$

Do we really need to retain the entire sample?

Can we replace the data by a lower-dimensional object without losing information about  $\theta$ ?

This leads to the theory of sufficient statistics.



## Takeaways

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1. A statistical model is a family of probability distributions.
2. Parametric models use a finite-dimensional parameter.
3. Nonparametric models allow much richer, often infinite-dimensional, parameter spaces.
4. Exponential families provide a powerful mathematical structure connecting likelihoods, moments, and sufficient statistics.
5. Location-scale families separate location and scale from distributional shape.

**Next lecture: Statistics, Sufficiency, Completeness, and Decision Theory**